

PYTHON

| INTRODUCTION

- ✓ Python is widely-used, high-level, interpreted, General purpose programming Language.
- ✓ Created by Guido van Rossum. He began working on python in the late 1980's and the first version of python was released in 1991 as 0.9.0.
- ✓ It is free and open-source.

| Application Based on PYTHON

- ✓ Web Development
- ✓ Software Development
- ✓ Game Development
- ✓ AI and Machine Learning and so on.

| Real-Time Applications

- ✓ Instagram
- ✓ Microsoft
- ✓ Yahoo
- ✓ Google
- ✓ Spotify
- ✓ Netflix
- ✓ Uber and so on.

| Characteristics of python

- ✓ Python is platform independent and so python code can be used for any operating systems like Windows, Unix, Linux, and Mac.
- ✓ Python is interpreted so doesn't need to compile, Python code automatically converts the source code into byte code internally, and the code is executed line by line.
- ✓ Python is simple so easy to understand and code.
- ✓ Python has a fewer lines of code.
- ✓ Python can be treated like either Object oriented, procedural and Functional ways.
- ✓ The biggest strength of python is, it has huge collection of standard libraries.
- ✓ It has a user-friendly data structure.

|Development Environment

- ✓ Need to download and install python from official website.
- ✓ We can create python file using text editors and IDEs.

|Input and Output Statements

- ✓ To perform a i/o tasks we have two build-in functions available in python.

|Input

- ✓ **Input ()** function is used to get input from the user, in python.
- ✓ Ex:

```
name = input ("Enter your name: ")
```
- ✓ Python takes all the inputs as string by default, to convert the input to int or float we must use the methods respectively.
- ✓ Ex:

```
number = int (input ("Enter Number:"))
```

|Output

- ✓ For displaying the output, the **Print ()** method is used.
- ✓ Ex:

```
Print ("Welcome to Techvolt Software")
```

|Output Formatting

- ✓ If we want to format our output, then we can use the method **str. format ()** like.
- ✓ Ex:

```
x = 5; y = 10  
print ('The value of x is {} and y is {}'.format (x, y))
```

|Statement

- ✓ Instruction that a python interpreter can execute are called **statements**.
- ✓ Ex: a = 1 is an assignment statement.

|Indentation

- ✓ Most of the programming languages like C, C++, and Java use brace {} to define a block of code. Python uses indentation.
- ✓ A code block (body of a function, loop, etc.) starts with indentation and ends with the first unindented line.

|Comments

- ✓ Comments are very important while writing a program. They describe what is going on inside a program.
- ✓ In Python, we use the hash (#) symbol to start writing a comment.
- ✓ Comments are for programmers to better understand a program. Python Interpreter ignores comments.
- ✓ Ex:

```
#This is a comment
#print out Hello
print('Hello')
```

|Variable

- ✓ Variable is a container used to store data, that can be changeable.
- ✓ We don't have to explicitly define the variable type, because python automatically identifies the type of the variable after assigning a value to it.
- ✓ Declaring and Assigning a value to variable
- ✓ Ex:

```
Text = "Welcome to Techvolt"
number = 10
```

- ✓ Changing the values to the variable
- ✓ Ex:

```
name = "Welcome to Techvolt"
print(name)
name = "Hi, Hello"
print(name)
```

- ✓ Assigning multiple values to multiple variable
- ✓ Ex:

```
a, b, c = 5, 3.2, "Hello"
print(a)
print(b)
print(c)
```

- ✓ Assigning same value to multiple variable
- ✓ Ex:

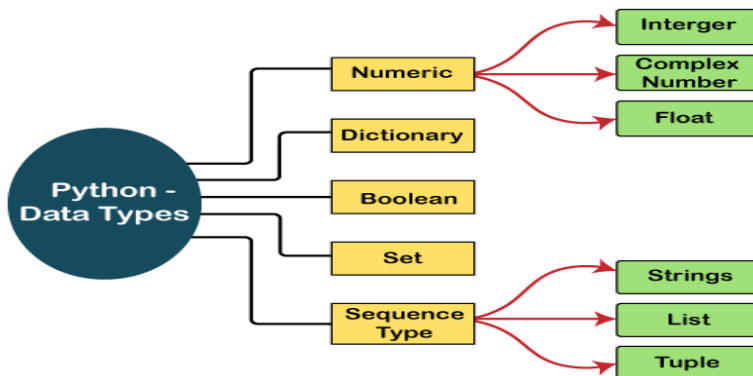
```
x = y = z = "same value"
print(x)
print(y)
print(z)
```

|Constant

- ✓ A constant is a type of variable whose value cannot be changed.

|Data types

- ✓ Every value in Python has a datatype.
- ✓ Since everything is an object in Python programming, data types are classes and variables are instance (object) of these classes.
- ✓ **type ()** function is used to know which class a variable or value belongs to.
- ✓ **isinstance ()** function is used to check if an object belongs to a class.



|Numbers

- ✓ Integers, floating point numbers and complex numbers comes under Python numbers category.
- ✓ They are defined as `int`, `float` and `complex` classes in Python.
- ✓ Ex:

```
a = 5
print (type(a))
a = 2.0
print (type(a))
a = 1 + 2j
print (type (a))
```

|List

- ✓ List is an ordered sequence of items.
- ✓ All the items in a list do not need to be of the same type.
- ✓ Items separated by commas are enclosed within bracket `[]`.
- ✓ Ex:

```
a = [1, 2.2, "python"]
```

|Tuple

- ✓ Tuple is an ordered sequence of items same as a list.
- ✓ The only difference is that tuples are immutable. Tuples once created cannot be modified.
- ✓ It is defined within parentheses `()` where items are separated by commas.
- ✓ Ex:

```
a = (5, "program", 1 + 3j)
```

|String

- ✓ String is sequence of Unicode characters.
- ✓ We can use single quotes or double quotes to represent strings.
- ✓ Multi-line strings can be denoted using triple quotes, `'''` or `"""`.
- ✓ Ex:

```
s = "This is a string"
s = """A multiline string"""
```

|Set

- ✓ Set is an unordered collection of unique items.
- ✓ Set is defined by values separated by comma inside braces `{}` Items in a set are not ordered.
- ✓ Ex:

```
a = {5, 2, 3, 1, 4}
```

|Dictionary

- ✓ Dictionary is an unordered collection of key-value pairs.
- ✓ In Python, dictionaries are defined within braces `{}` with each item being a pair in the form of `key: value` pair. Key and value can be of any type.
- ✓ Ex:

```
d = {name: "viveka"}
```

|Type Conversion and Type Casting

- ✓ The process of converting the value of one data type (integer, string, float, etc.) to another data type is called type conversion.
- ✓ Python has two types of type conversion.
 1. Implicit Type Conversion
 2. Explicit Type Conversion

|Implicit Conversion

- ✓ In Implicit type conversion, Python automatically converts one data type to another data type.
- ✓ This process doesn't need any user involvement.
- ✓ Ex:

```
a = 5
b = 10.5
c = a + b
```

|Explicit Conversion

- ✓ Users can convert from one data type to another data type.
- ✓ We can use the functions like `int ()`, `float ()`, `Str ()` to perform explicit type conversion.
- ✓ Ex: `a = str (5).`